

BHARAT REFINERIES LIMITED
MECHANICAL INSPECTION DEPARTMENT

IN-SERVICE INSPECTION REPORT

Report No. : IR-2026-0731
Date : 31 July 2026
Inspector : R. Krishnan, Level II UT
Unit : Crude Distillation Unit 2
Procedure : SOP-MECH-014 Rev 4

(SYNTHETIC DOCUMENT - created for system testing.)

1. EQUIPMENT PARTICULARS

Equipment Tag : V-204
Description : Overhead Reflux Accumulator
Registration No. : PV-CDU2-0044
Year of Commissioning : 2016
Nominal Thickness : 12.0 mm
Corrosion Allowance : 3.0 mm
Minimum Required Thick : 7.5 mm
Design Pressure : 16 bar g
Design Temperature : 205 deg C
Operating Pressure : 12 bar g
Operating Temperature : 138 deg C

2. PREVIOUS INSPECTION

Previous Inspection Ref : IR-2023-0512
Previous Inspection Date: 28 July 2023
Thickness Recorded : 10.4 mm

3. THICKNESS SURVEY - CURRENT

Location	Reading (mm)
Shell Course 1 N	9.4
Shell Course 1 E	9.2
Shell Course 1 S	9.6
Shell Course 1 W	9.3
Bottom Head	9.8
Top Head	10.1

Governing (minimum) reading: 9.2 mm at Shell Course 1 East

4. VISUAL AND NDT FINDINGS

- 4.1 General external condition satisfactory. Insulation cladding intact.
- 4.2 Localised external corrosion observed at the shell-to-skirt junction over an area of approximately 300 mm x 150 mm. Pitting depth not exceeding 0.8 mm.
- 4.3 Magnetic particle inspection of the inlet nozzle weld N1 showed no detectable linear indications.
- 4.4 Pressure relief valve PSV-204A last tested 12 March 2026, set pressure 16 bar g, certificate PSV/2026/0311 refers.
- 4.5 The vessel nameplate is legible and matches the registration record.

5. OBSERVATIONS REQUIRING ACTION

5.1 The governing thickness reading has fallen to 9.2 mm against a minimum required thickness of 7.5 mm.

5.2 External corrosion at the shell-to-skirt junction requires surface preparation and re-coating at the next available shutdown.

6. INSPECTOR'S REMARKS

The vessel remains fit for continued service at its current operating pressure. The rate of wall loss since the previous survey requires evaluation against clause 4.4 of SOP-MECH-014 before continued operation is approved.

Signed: R. Krishnan

Date : 31 July 2026